

ECE 2210: Python Programming for ECE

HW1 – Arithmetic Operators

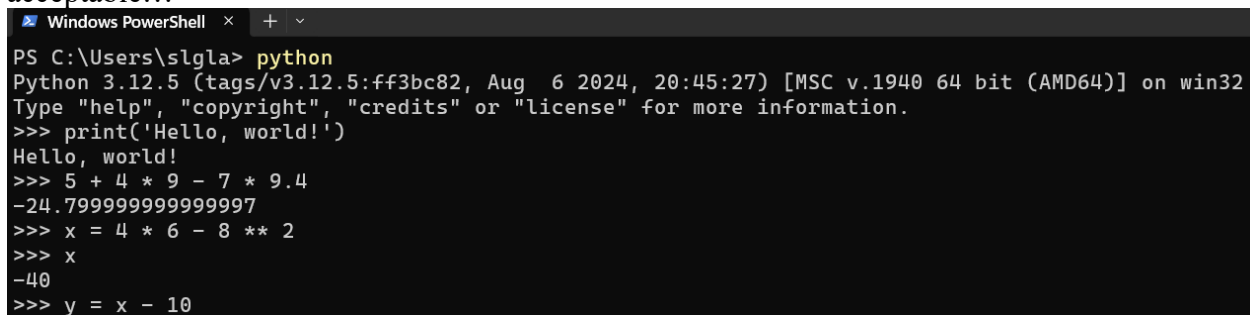
Start the Python 3 in the *interactive mode*.

- (20 pts) Arithmetic expressions and data types.
 - What is the result of “10 / 16”?
 - What is the data type of the result of “10 / 16”? *Hint*: you can pass an expression as the input to a function.
 - What is the data type of the result of “10 / 10”?
 - What is the data type of the result of “5 * 6”?
 - What is the data type of the result of “5.0 * 6”?
- (20 pts) How many seconds are there in 42 minutes 42 seconds?
- How many miles are there in 10 kilometers? *Hint*: there are 1.61 kilometers in one mile.
- If you run a 10-kilometer race in 42 minutes 42 seconds, what is your average pace (time per mile in **seconds**)?
- What is your average speed in miles per hour in the last question?

Submission: The submission is a PDF file called **HW1.pdf**, which should contain **two** screenshots of the terminal showing your interaction with Python interpreter. Specifically,

- The 1st screenshot (named Problem 1) shows your answers to Problem 1, which contains the expression/statement entered (for each subproblem) with the evaluation/execution result displayed in the very next line. Or, you can assign the evaluation result of the expression in a variable¹ and then display the variable value in the following lines (see Figure 1). **ONLY ONE** expression is allowed for each subproblem.
- The 2nd screenshot (named Problem 2 – Problem 5) shows your answers to Problem 2 ~ Problem 5, which contains the arithmetic expression entered (for each problem) with the evaluation result displayed in the very next line or the line after if you choose to assign the result in a variable. **ONLY ONE** expression is allowed for each problem.

A screenshot example is given below. **Note** that pictures taken with cellphones are not acceptable!!!



```
Windows PowerShell x + v
PS C:\Users\slgla> python
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print('Hello, world!')
Hello, world!
>>> 5 + 4 * 9 - 7 * 9.4
-24.799999999999997
>>> x = 4 * 6 - 8 ** 2
>>> x
-40
>>> y = x - 10
```

Figure 1 This screenshot contains the results of executing one statement and evaluating three arithmetic expressions.

¹ The reason for temporarily storing the value in a variable is it can be used later in the following problems.